Can a Madrid + Sevilla price model serve Barcelona?

Team ISD · Group Assignment 2 · Airbnb Spain on Databricks Free Edition

The task: price suggestions for two clients, then a third

City A · training

Madrid

Snapshot 2026-09-17
23,109 listings

City B · training

Sevilla

Snapshot 2026-09-25
8,375 listings

City C · new client

Barcelona

Snapshot 2026-09-17
15,236 listings

- Week 1: EDA, medallion lakehouse, one job with the city as a parameter, a price model beaten against a fixed baseline, an AI/BI dashboard and a Genie agent, for Madrid and Sevilla.
- Week 2: run the frozen pipeline and model on Barcelona, fix what breaks, and explain the gap.

Analysis: missing prices mark inactive listings, and two columns are the price in disguise

18.7% / 7.8%

of Madrid / Sevilla listings have no price

99%

of inactive Madrid listings (no availability, no recent reviews) have no price

< 1%

priced above 1000 EUR: mostly parked or mistyped prices, flagged not dropped

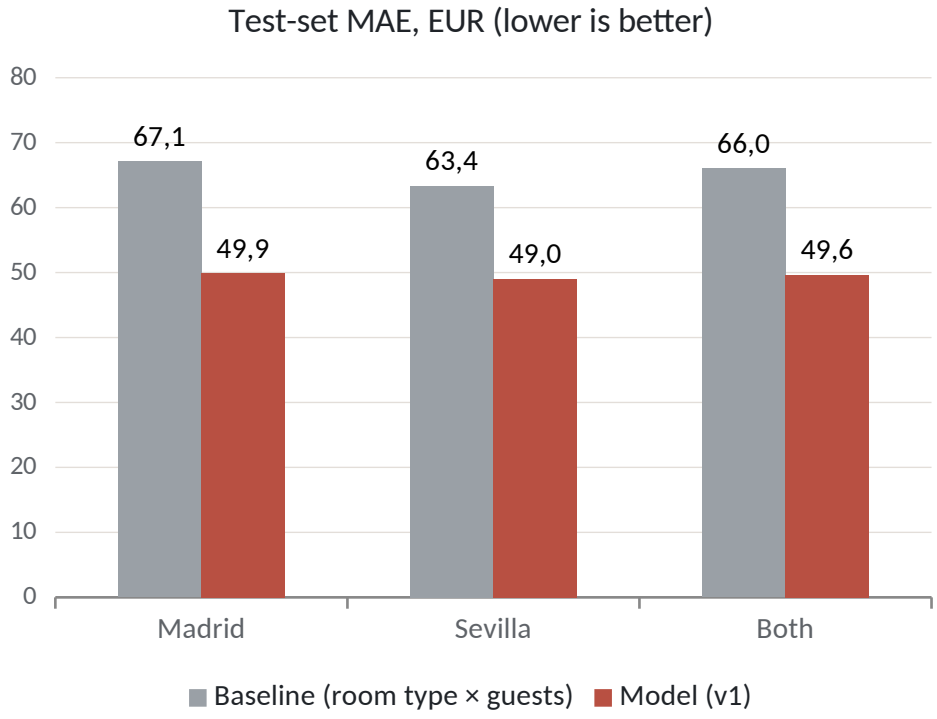
Column	Relation to price	Verdict
price_quote_price_per_night	equals price for 100% of rows	leak
estimated_revenue_l365d	= price × occupancy (ratio 1.000)	leak
price_quote_total / raw / dates	the same quote	leak
estimated_occupancy, availability_*	outcomes of the price	excluded
reviews, review scores	listing history	allowed

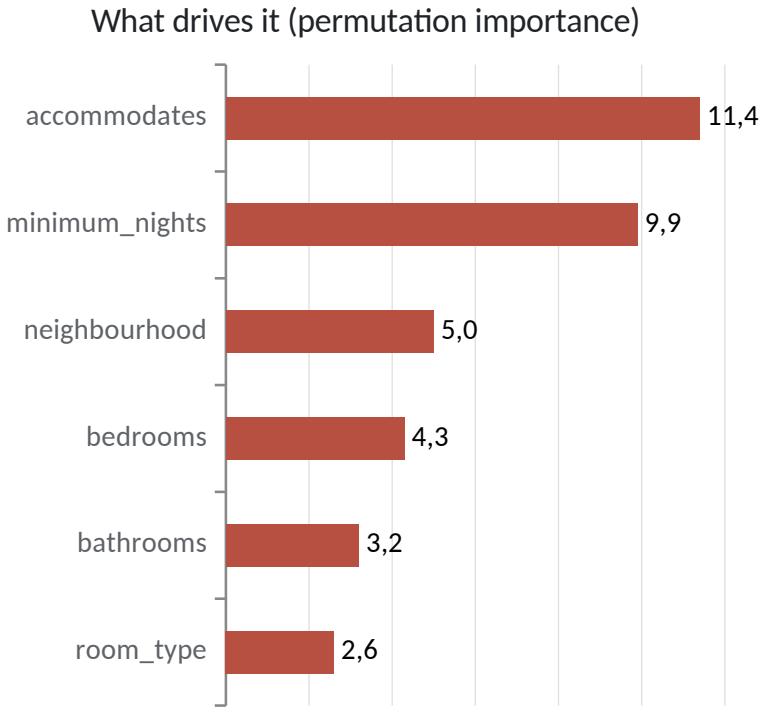
Architecture: one parameterised job turns a raw file into predictions



- Job airbnb_pipeline: for each city → ETL; train on train_cities (optional); for each city → score.
- The city is a parameter; everything city-specific lives in config/cities.yml. Adding Barcelona was one config entry.
- Every write replaces only its own city's rows, so cities re-run independently. Model registered in Unity Catalog (price_model@prod).

The model beats the baseline by a quarter in both training cities





The new client: Barcelona

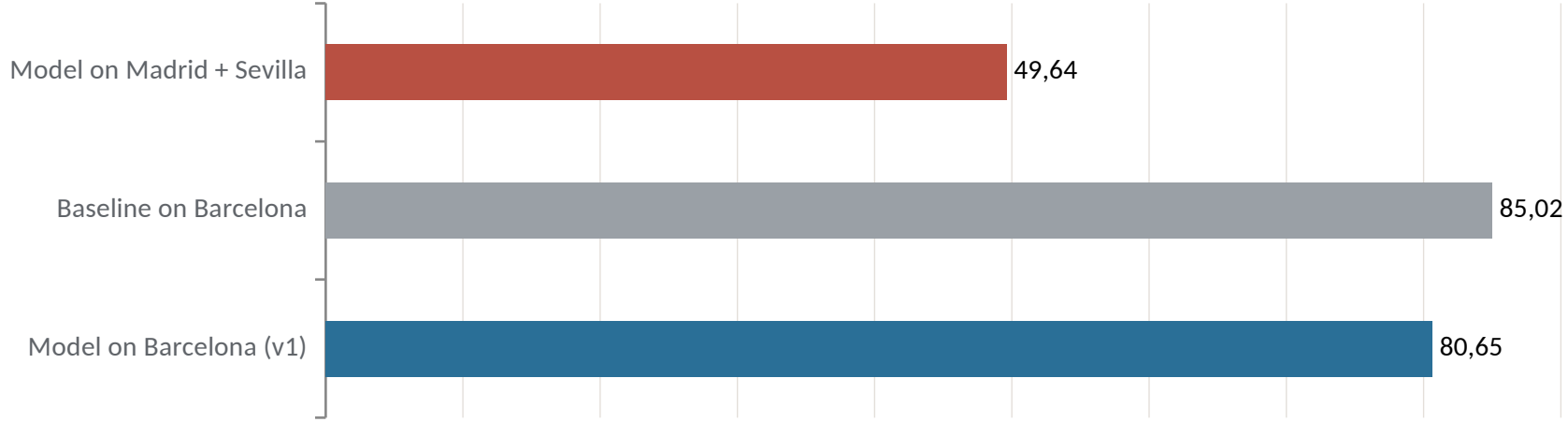
“Your pipeline takes the city as a parameter, right?”

First run: nothing crashed, and the model barely beat the baseline

1 min 52 s
job succeeded; download from Inside Airbnb worked

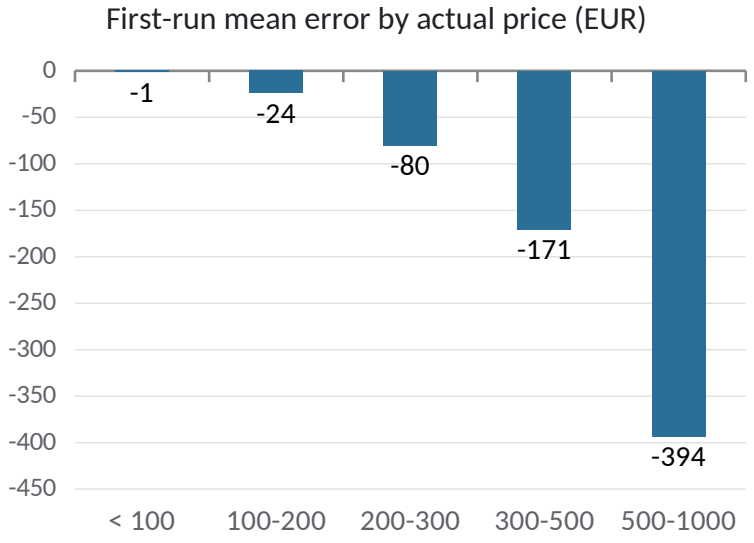
15,236
rows in bronze, silver and features = source count

68 of 68
Barcelona neighbourhoods unknown to the model, and no warning



Why it did not transfer: a price level, a stay mix and neighbourhoods it never saw

	Madrid	Sevilla	Barcelona
Median price, stays < 30 nights	157	154	**246**
Minimum stay ≥ 30 nights	12.5%	1.8%	**33.5%**
Neighbourhoods the model has seen	all	all	**none**
Price spread (IQR / median)	0.84	0.63	**0.98**



The model predicts Madrid prices: it under-prices every expensive Barcelona listing.

What we changed: one pipeline fix and one model change

Pipeline fix

Scoring now records unseen_<column>_share per city in model_metrics and warns above 20%.

Why: the pipeline should have told us it was scoring a city with 100% unknown neighbourhoods.

Model change

Retrain on the training split of Madrid + Sevilla + Barcelona (job parameter train_cities, no code change). Model v2.

Why: the model had never seen Barcelona's price level, stay mix or neighbourhoods.

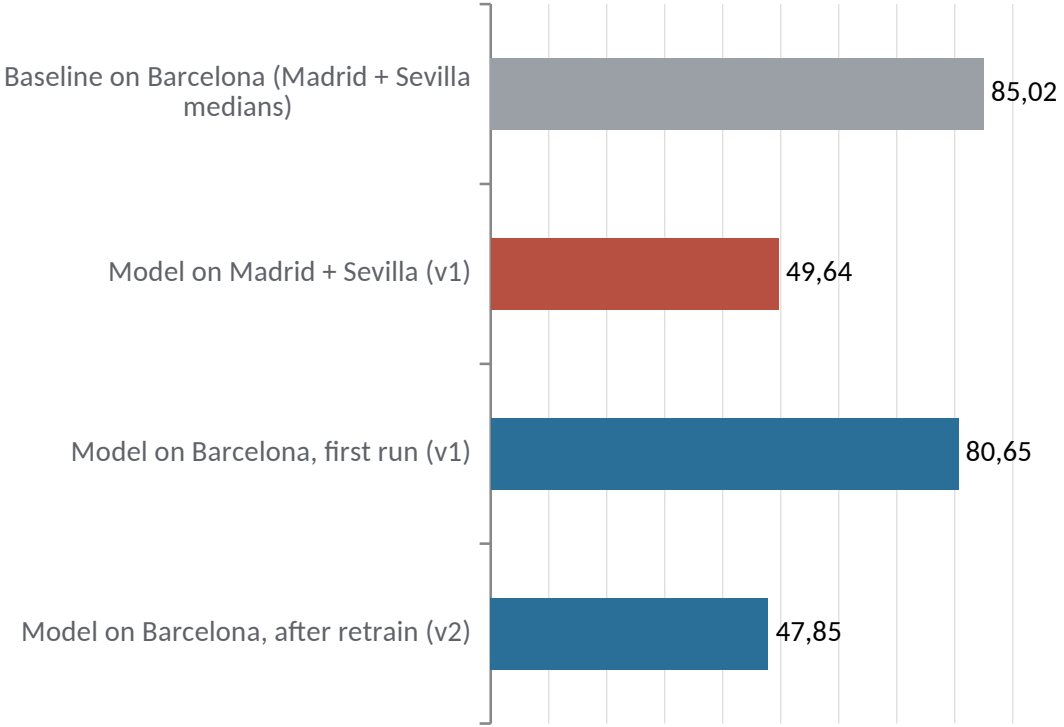
Considered, not adopted

Drop neighbourhood (zero-shot 67.4 EUR); a separate price cap for Barcelona.

Why not: with Barcelona data the feature helps again; one cap keeps cities comparable.

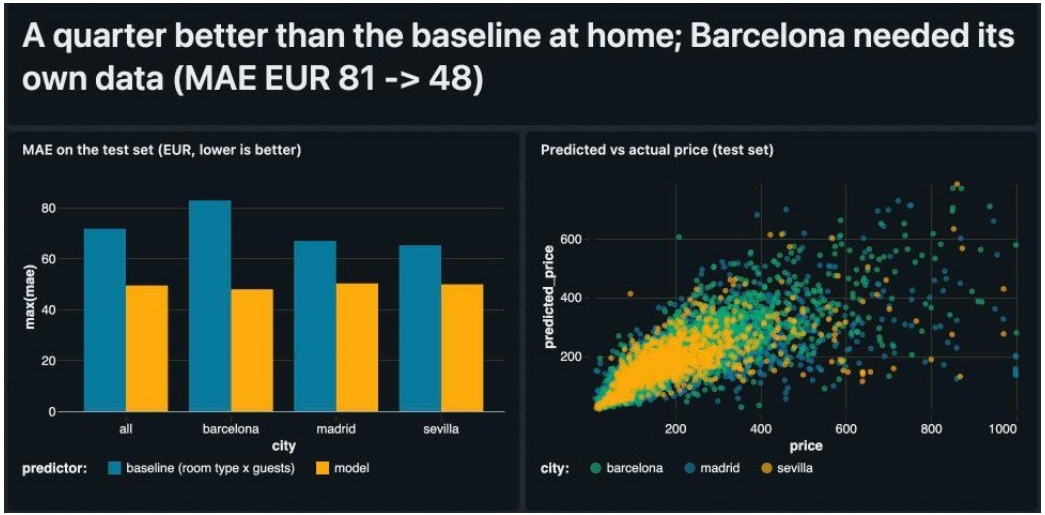
Every change is a commit or a job run in the repository; the Barcelona test split was never used for training or for choosing between options.

Barcelona's own data cuts the error from 81 to 48 EUR



- Training cities after retrain: Madrid 50.15, Sevilla 49.80 EUR (from 49.89 and 49.01).
- Barcelona's own-data baseline is 81.16 EUR: the city is harder, its prices are more spread out.
- Including prices above 1000 EUR: first run 122.99, after 81.83 EUR.

Dashboard and Genie: every page title is a finding; Genie knows what “City A” means



- Three pages: City overview, Madrid vs Sevilla, Model (Barcelona added after the C run).
- Genie tricks: letters → city names, the sign of error spelled out, test-split filter by default, minimum group size, Unity Catalog column comments, the brief's three questions as example SQL.
- All three example questions answered correctly (174 EUR; Alfalfa 938; El Viso −105).

Challenges, and who did what

- Free Edition limits: no bundles in the UI, restricted internet. Fix: the job is created by a notebook; the Volume fallback for raw files.
- 19-digit listing ids broke the fallback cast on older Spark clients; caught in review before the first Databricks run.
- Two 2026 columns leaked the price; found only by checking ratios, not correlations.

Team member	Contribution
(name)	EDA and leakage review
(name)	Pipeline and Databricks job
(name)	Model and evaluation
(name)	Dashboard and Genie
(name)	City C run and report

The pipeline is a product. The model is per city.

One config line took Barcelona from raw file to predictions without a failure. A new client still needs its own training data, and now the pipeline says when it does not have it.